

Vandad Davoodnia

✉ vandad.davoodnia@gmail.com ☎ (+1) 647-472-6419 🏠 Toronto, ON

in Vandad-Davoodnia 🎓 Google Scholar 🌐 Website

PhD graduate with 2 years of industry experience in machine learning and computer vision. 12+ years of professional programming experience and 3+ years of data science research experience with 14 publications and reviewer of 25+ papers. Interested in computer vision and natural language processing and related research in generative modeling and self-supervised learning.

Technical Skills

Programming	Python, C/C++, C#, Java, LISP, MATLAB, JavaScript, TypeScript, HTML, CSS, PHP
Libraries	PyTorch, TensorFlow, FastAPI, Scikit-learn, SciPy, OpenCV, MLFlow, SDL, GTK+
Technologies	Git+Github Actions, Gitlab+CI/CD, Docker, SLURM, Kubernetes, SQL, MongoDB
Graphics	Unity 3D, Blender+API, Motion Builder+API, 3DsMax
Familiar with	LangChain, RAG methods, JIRA, MLFlow, AWS Bedrock/Lambda/SageMaker/S3

Professional Experience

Ubisoft LaForge

Toronto, ON, Canada

Research and Development Programmer, Part-time

May 2022 - Jan 2024

- **AI Model Development**
Developed a markerless MoCap application, including human tracking, re-identification, camera calibration, pose estimation, and 3D mesh reconstruction, for low-cost and portable animation prototyping
- **AI Model Deployment**
Deployed the Docker image of the markerless motion capture application in Kubernetes cluster, including API access with FastAPI and GUI with Gradio in Python
- **Pushing State of the Art**
Obtained competitive results compared to top industry competitors (*XSens, MoveAI*) while achieving top performance among academic models by 47% error reduction on out-of-distribution data
- **Academic Publications**
Published a paper on 3D human pose estimation in **ECCV 2024** and a paper on human 3D mesh estimation in **ECCV 2024 workshops**
- **Generative Model Exploration**
Explored generative models (VAE, VQ-VAE, GAN, Denoising Diffusion) for developing a pose embedding and facilitating human motion generation in PyTorch
- **Synthetic Data Generation**
Generated synthetic data using Python in Blender and MoBu for training pose estimation models
- **3D Kinematic Body Mesh Modeling**
Developed a parametric 3D body model with blend shapes in PyTorch to bridge the gap between internal motion capture datasets and academic research and expedite future research projects

Brain Engineering Center, IPM

Tehran, Iran

Research Scientist, Part-time

Sept 2015 - July 2018

- **Conventional ML Model Development**
Developed a neuro-fuzzy model (ANFIS) and trained it for biomass behavior prediction using PSO in MATLAB as a part of academic paper publication
- **Data Collection**
Assisted with data collection efforts and cleaned physiological signals using variety of statistical methods, such as PCA, t-SNE, GMM, and K-means in Python and MATLAB

- **Software Development**

Developed a software for Blackrock Microsystems Inc. to enable synchronized video and multi-neuron recordings using Qt and OpenCV in C++

- **Mobile App Development**

Developed Eye Speed app (iOS/Objective-C, Android/Java) for glaucoma diagnosis as a part of academic paper publication

- **Time-series Analysis**

Extracted features from physiological time-series and performed causal and correlational analysis between different sources and variables

- **Statistical Analysis**

Developed a neural response tuning method as a part of academic paper publication

- **Physiological Lab Setup**

Managed the lab's equipment and implemented a real-time experimental control software in MATLAB for primate training

- **Medical Image Analysis**

Performed MRI segmentation and provided 3D visualizations for surgical decision-making using MATLAB

Freelancer

Tehran, Iran

Software Developer

Oct 2012 - July 2018

- **Computer Vision & Robotics**

Developed a robot and its interface for object and path tracking using OpenCV in C++

- **Business Contracts**

Implemented software for simulating Iran's transit roads and rails network using SQL and Transcad/GISDK to expedite decision making

- **Software Development**

Programmed over 20 games, apps, and simulation programs in C, C++, Java, MATLAB, and Python

Education

Queen's University

Kingston, Canada

Ph.D. Candidate, Electrical and Computer Engineering (A⁺)

Sept 2018 - June 2024

Dissertation: Estimating Human Pose from Pressure and Vision Data ([Link](#))

- **Multi-modal LLMs**

Developed a pose-aware LLM for accurate body posture understanding using a synthetic dataset

- **Publication Record**

Authored 5 papers on pose estimation using methods such as statistical machine learning, self-supervision, and domain adaptation, leading to publications in top venues, *e.g.*, ECCV and ICASSP

- **Collaborative Research on Diverse Applications**

Co-authored 5 papers on supervised & semi-supervised EEG representation learning, the impact of e-textiles on drivers' experience, cancer prediction from medical features, and irony detection in tweets

- **Grant Acquisition & Industry Partnership**

Secured \$30,000 Mitacs internship grant supporting 8 months of research at Ubisoft Toronto Inc.

- **Vision Research Experience**

Explored consistency regularization, self-supervised learning, and differential rendering for 3D human pose, appearance, and body shape representation learning from monocular images

- **Data Collection**

Assisted with data collection to study the correlation between drivers' experience and sensors such as ECG, face camera, pressure, and pupil size

- **Cloud Infrastructure Management**

Managed lab's GPU servers and cloud services, ensuring a secure and smooth user experience

Sharif University of Technology

Tehran, Iran

B.Sc., Electrical Engineering - Bioelectronics (B⁺)

Sept 2012 - June 2017

Honors

- **Ranked 4th** in the nationwide university entrance exam (Concours) for B.Sc. admission among 500,000+ high school graduate examinees 2012

Publications

Conferences

- **Davoodnia V.**, Ghorbani S., Carbonneau M. A., Messier A., Etemad A., "UPose3D: Uncertainty -Aware 3D Human Pose Estimation with Cross-View and Temporal Cues", *ECCV*, 2024
- **Davoodnia V.**, Ghorbani S., Messier A., Etemad A., "SkelFormer: Markerless 3D Pose and Shape Estimation using Skeletal Transformers", *ECCV Workshops*, 2024
- Khorsandi P. M., Jones L., **Davoodnia V.**, Lampen T. J., Conrad A., Etemad A., Nabil S., "FabriCar: Enriching the User Experience of In-Car Media Interactions with Ubiquitous Vehicle Interiors using E-textile Sensors", *ACM DIS*, 2023
- **Davoodnia V.**, Etemad A., "Human Pose Estimation from Ambiguous Pressure Recordings with Spatio-temporal Masked Transformers", *IEEE ICASSP*, 2023
- **Davoodnia V.**, Ghorbani S., Etemad A., "In-bed Pressure-based Pose Estimation using Image Space Representation Learning", *IEEE ICASSP*, 2021
- **Davoodnia V.**, Etemad A., "Identity and Posture Recognition in Smart Beds with Deep Multitask Learning", *IEEE SMC*, 2019
- Sarkar P., **Davoodnia V.**, Etemad A., "Computer-Aided Diagnosis using Class-Weighted Deep Neural Networks", *IEEE ICMLA*, 2019

Journals

- Zhang G., **Davoodnia V.**, Etemad A., "Parse: Pairwise alignment of representations in semi-supervised EEG learning for emotion recognition", *IEEE TAFEC*, 2022
- **Davoodnia V.**, Ghorbani S., Etemad A., "Estimating Pose from Pressure Data for Smart Beds with Deep Image-based Pose Estimators", *Springer APIN*, 2021
- **Davoodnia V.**, Slinowsky M., Etemad A., "Deep Multitask Learning for Pervasive BMI Estimation and Identity Recognition in Smart Beds", *Springer AIHC*, 2020
- Zhang G., Sepas-Moghaddam A., **Davoodnia V.**, Zhang Y., Etemad A., "Classification of Hand Movements from EEG using Attention-based LSTM", *IEEE Sensors Journal*, 2019
- Aghbashlo M., Tabatabaei M., Nadian M. H., **Davoodnia V.**, Soltanian S. "Prognostication of Lignocellulosic Biomass Pyrolysis Behavior using ANFIS Model Tuned by PSO Algorithm", *Fuel*, 2019
- Fayyaz Z., Bahadorian M., Doostmohammadi J., **Davoodnia V.**, Khodadadian S., Lashgari R., "Multifractal Detrended Fluctuation Analysis of Continuous Neural Time Series in Primate Visual Cortex", *Neuroscience Methods*, 2019
- Zhao, L., Sendek, C., **Davoodnia, V.**, Lashgari, R., Dul, MW., Zaidi, Q., Alonso, JM., "Effect of Age and Glaucoma on the Detection of Darks and Lights", *IOVS*, 2015

Services

Area Chair/ Program Committee Member

- AAAI R2HCAI Workshop 2023

Reviewer

- ICASSP, ICLR, IEEE TAI, Springer APIN 2025
- NeurIPS, IEEE SMC, ICASSP, IEEE TAI, Springer APIN 2024
- IEEE TAI, IEEE JBHI, Springer APIN, AAAI R2HCAI Workshop (**Best Reviewer Award**) 2023
- IEEE Sensors Journal, Springer APIN, IEEE TAI 2022
- ACII, Springer APIN 2021
- IEEE TNSRE, CVPR, IEEE SMC, ICASSP 2020
- IEEE Sensors Journal, ACM SAP, IEEE SMC 2019

Teaching Assistant Roles

Computer Architecture, Prof. Naraig Manjikian, Queen's University 2019 & 2021 & 2022 & 2023

- Graded exams and assisted in laboratory sessions with Assembly

Operating Systems, Prof. Thomas Dean, Queen's University 2020

- Graded exams and assisted in laboratory sessions with C++

Artificial Intelligence and Interactive Systems, Prof. Ali Etemad, Queen's University 2018 & 2020

- Graded exams and assisted in laboratory sessions with MATLAB and Python
- Developed course materials from "Artificial Intelligence - A Modern Approach" by Russell

Related Academic Courses

Machine Learning for Natural Language Processing, Prof. Xiaodan Zhu (Ph.D. A⁺) 2019

- Created an NLP model for detecting irony on Tweets, achieving 10% improvement over the top competitor

Biological Signal Analysis, Prof. Evelyn Morin (Ph.D. A⁺) 2019

- Developed an ML solution for predicting heavily loaded lifts from EMG signals with less than 100g error

Advanced Research Human-Computer Interactions, Prof. Nick Graham (Ph.D. A⁻) 2018

- Developed a hand movement tracking software using 3DsMax, Vicon MoCap, and Unity 3D

Wearable and IoT Computing, Prof. Ali Etemad (Ph.D. A⁺) 2018

- Developed a deep-learning model for user and posture detection in smart beds, resulting in a publication

Computational Intelligence, Prof. Sepideh Hajipour (B.Sc.) 2016

- Implemented a neural network for BCI2003 competition using feature extraction and feature selection